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Problem Statement ID - 26248

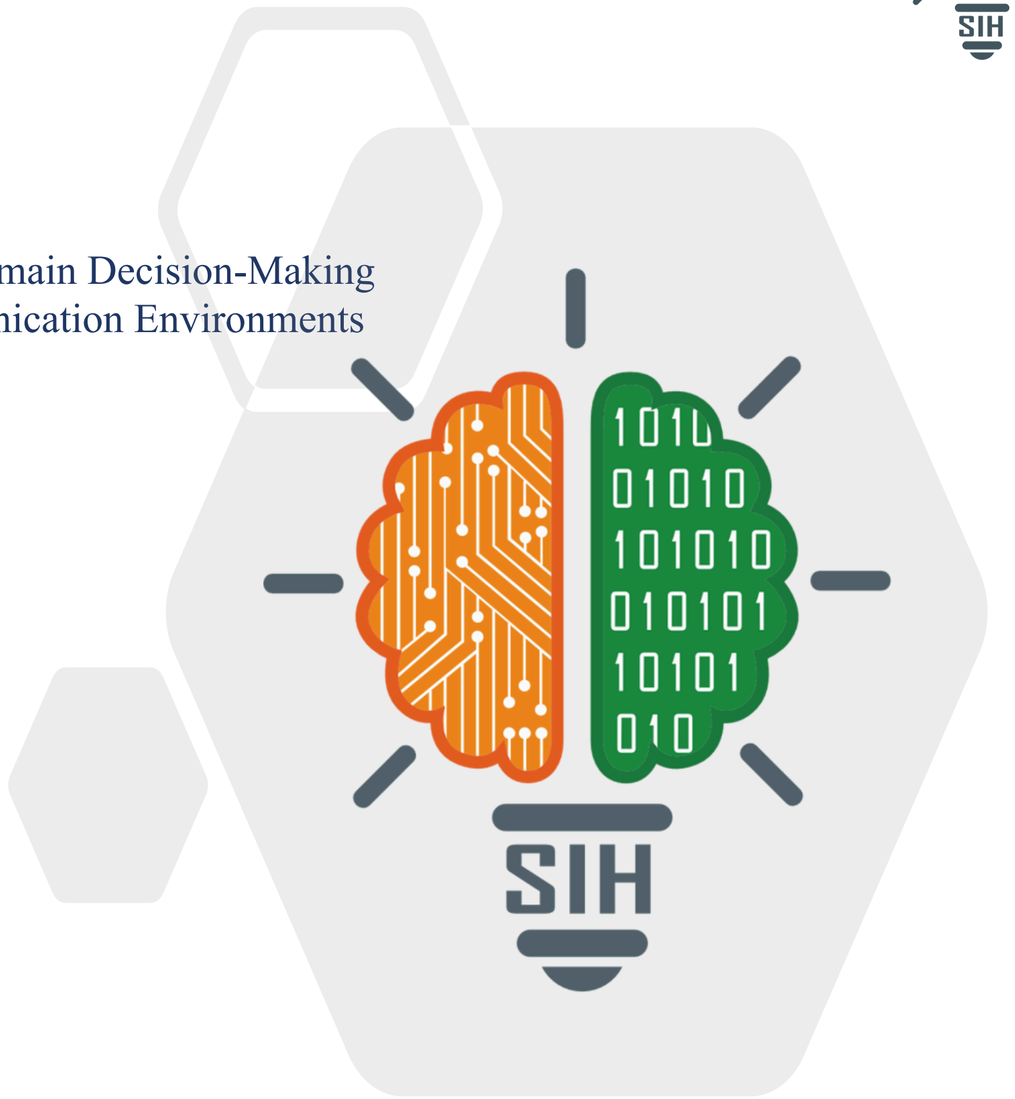
Problem Statement Title - SituSim: Immersive Multi-Domain Decision-Making Trainer for Degraded Communication Environments

Theme - Smart Automation

PS Category - Software

Team ID - 175991

Team Name - Algoholic1



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SituSim

Immersive Multi-Domain Decision-Making Trainer

Training Smarter Decisions When Communication Fails

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THE PROBLEM

Military teams need to make coordinated decisions across domain even when communication networks are degraded or unavailable.

Communication loss
Radio and data links may fail during critical missions.

Siloed Training
Domain-specific drills limit joint decision practice.

Delayed Decisions
Incomplete information slows mission response.

Limited Realism
Training may not reproduce contested communication conditions

OUR SOLUTION

An immersive, multi-domain training simulator that builds decision-making and coordination skills in degraded communication environments.

Immersive Simulator
Runs in any browser: live map, fog of war, time pressure.

Multi-Domain
Integrates cross-domain decisions and coordination.

Comms Degradation
Simulates link failures and limited information.

Decision Evaluation
Reviews response time, choices and coordination.

INNOVATION & UNIQUENESS

Comms-Aware Training
Practice decisions under simulated communication loss.

Information gap vs reasoning gap
Tells whether to fix the comms or the reasoning.

See what they saw
Replay any trainee's screen; decisions judged against what they had.

Talking has a cost
Detectability and enemy direction-finding makes comms discipline part of the exercise.

Offline-Ready
Supports training when external connectivity is unavailable.

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TECHNICAL APPROACH

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1 SCENARIO SETUP



- Instructor configures mission context.
- Defines team roles, objectives and assets.
- Sets scenario injects and timeline.

2 MULTI-DOMAIN SIMULATION



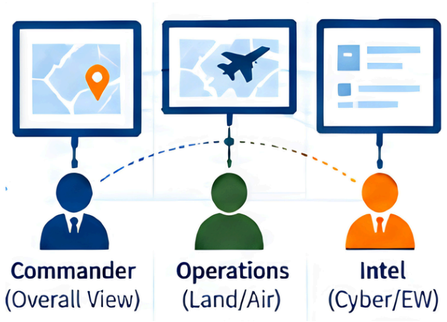
- Simulates connected multi-domain events.
- Land, air, cyber and electronic warfare scenarios.

3 COMMUNICATION DEGRADATION



- Injects configurable delay, dropout, missing data and contradictory reports
- Differentiated information across team members

4 TEAM DECISIONS



- Role-based views for each team member
- Communicate and coordinate under changing information
- Make time-critical decisions as a team

5 AFTER-ACTION REVIEW



- Review individual and team decisions
- See what information was available at each point
- Export detailed exercise report

KEY CAPABILITIES

Scenario Control

Flexible mission setup and inject configuration

Degraded-Info Simulation

Configurable delays, dropouts, incomplete and conflicting data

Team Coordination

Role-based views and real-time collaboration

Decision Analytics

Detailed AAR with timelines and rationale

TECH STACK

FastAPI, python, websockets, SQLite, PostgreSQL, Jinja, uvicorn, HTML, CSS, JS, SVG, Playwright, Render

Github
YT Video
Live Prototype

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FEASIBILITY AND VIABILITY

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ANALYSIS OF THE FEASIBILITY OF THE IDEA

1. TECHNICAL FEASIBILITY

A web app on one Python server; trainees need only a browser. Works without internet. Already built and tested end to end with 4 roles.

2. ECONOMIC FEASIBILITY

Built with free, open-source tools and everyday hardware. Once made, a scenario can be reused many times, so training costs far less than live exercises.

3. OPERATIONAL FEASIBILITY

Trainers can set up a drill in a few clicks and replay it later to discuss mistakes. It works in low- or no-connectivity bases without changing the existing training workflow.

4. DATA FEASIBILITY

We start with made-up (synthetic) scenarios based on standard military procedures and open maps. No secret or classified data is needed to get started.

5. SCIENTIFIC VIABILITY

Based on how people really make decisions under pressure. Measures more than just speed: information gap vs reasoning gap, calibration (Brier score) and SAGAT-style freeze probes.

POTENTIAL CHALLENGES AND RISKS

NETWORK NOT REAL ENOUGH

Our fake jamming and delays may feel different from a real battlefield.

TOO MANY MOVING PARTS

Land, air, cyber and EW all affect each other, which is hard to model.

AUTOMATIC SCORING

An automatic score could misjudge a decision
.

LACK OF REAL DATA

We may not get real operational data, so scenarios could feel less real.

PEOPLE MAY RESIST

Trainers and trainees need time to get used to a new way of learning.

KEEPING DATA SAFE

Training scenarios can be sensitive and must be protected.

STRATEGIES FOR OVERCOMING THESE CHALLENGES

REALISTIC NETWORK TESTER

Adjustable delay, message loss and blackout settings, tuned using feedback.

BUILD IN SMALL BLOCKS

Scenarios are JSON files: new terrain, roles or injects need no new code.

TRANSPARENT SCORING

A rule-based shadow analyst; every finding cites event IDs; it supports the instructor's judgment, it doesn't replace it.

EXPORT-GUIDED SCENARIOS

Planned: scenarios written and refined with DSSC instructors after each trial.

FOG DIRECTOR & REPLAY

Adjusts the fog automatically from how the team is coping; replay any trainee's screen at any second.

PRIVATE & SECURE

Runs fully offline on the unit's own network; the instructor console is passcode-locked and a live exercise's review stays closed until it ends.

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IMPACT AND BENEFITS

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MARKET AND ADOPTION

GLOBAL MARKET

Military simulation and training:

- USD 13.39B (2024)
- USD 21.08B by 2030
- 7.85% CAGR

ROUTE TO ADOPTION

iDEX: Ministry of Defence channel for defence start-ups.

PROPOSED ADOPTION PATH

Pilot at DSSC Wellington → Army War College and MCTE Mhow → Civil disaster drills with NDMA, NDRF and SDRF.

IMPACTS

Train For the real Fight



Trains decisions under delay, loss and deception, not clean data.

Fairer Debriefs



Identifies whether mistakes occur due to communication gaps or reasoning errors, enabling targeted improvement.

Comms Discipline



Channel switching and emission control are measured, not assumed.

Dual Use



The same tool could run disaster-response drills where networks fail.

BENEFITS

AAR in Seconds



Built from the event log the moment the exercise ends.

No new Hardware



Works with one laptop and trainees' phones; no simulation centre.

Evidence, not Opinion



Logs trainee information delivery for repeatable analysis.

Calibration Training



Reduces overconfidence through probability training.

WHERE IT SITS AGAINST THE ALTERNATIVES

Information	`	SituSim
Information delivery	Complete and on time	Delayed, lost and contradicting, by design
Who knew what	Not recorded	Every delivery logged per trainee
AAR preparation	Manual notes afterwards	Generated automatically at the end
Equipment	**Staffed exercise set-up**	**One laptop and phones**

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RESEARCH AND REFERENCES

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3. Klein, G., Sources of Power: How People Make Decisions, MIT Press, 1998.
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OUR PROTOTYPE

